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RESEARCH

BDS-3 PPP-B2b and Galileo HAS tightly integrated real-time PPP

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Abstract
The BDS-3 PPP-B2b of China can offer BDS-3/GPS real-time precise point positioning (PPP) service in the Asia-Pacific region, while the Galileo High Accuracy Service (HAS) of the European Union can offer Galileo/GPS real-time PPP service worldwide. Their service areas have a geographical overlap in the Asia-Pacific region and there is potential for their interoperability in terms of supported satellite systems. In this study, a tightly integrated real-time PPP model by incorporating observations from complementary constellations of PPP-B2b and HAS (rather than combining their PPP results) based on respective stand-alone service for the Asia-Pacific region is constructed. For comparison, we employ the strategy with BDS-3/GPS from PPP-B2b (B2b C/G), the strategy with Galileo/GPS from HAS (HAS E/G), the strategy with BDS-3/GPS from PPP-B2b and Galileo from HAS (B2b C/G+HAS E), and the strategy with Galileo/GPS from HAS and BDS-3 from PPP-B2b (HAS E/G+B2b C). The datasets from 18 stations over a week are employed. The B2b C/G+HAS E strategy has the best performance, followed by B2b C/G and HAS E/G+B2b C strategies, while the performance of HAS E/G strategy is the worst. The B2b C/G+HAS E strategy can provide a convergence time (processed every two hours) of 9.4 min with a threshold of 20/40 cm in horizontal/vertical directions, and a positioning accuracy (processed every 24 h) of 4.2/3.2/9.1 cm in east/north/up directions. Therefore, benefiting from the increased satellite number and improved satellite geometry, the integration of PPP-B2b and HAS can significantly improve the real-time PPP performance compared with respective stand-alone service.

Keywords BDS-3 PPP-B2b · Galileo HAS · Real-time precise point positioning · Tight integration · Satellite communication

Introduction

Precise point positioning (PPP) technology enables users to achieve high-precision positioning with a single receiver through the joint use of code and carrier phase observations as well as precise satellite orbit and clock products. With the ongoing advancement of PPP technology and the rapidly increasing demand for real-time precise positioning, real-time PPP has become a research hotspot. The international Global Navigation Satellite System (GNSS) service (IGS) has been delivering real-time precise correction products via Internet, providing users with real-time service (RTS) that can achieve decimeter-to-centimeter positioning accuracy (Nie et al. 2018; Elsobeiey and Al-Harbi 2016). However, the implementation of RTS relies on a stable network communication environment, which restricts the application of real-time PPP technology in areas with insufficient network coverage, such as deserts and oceans. In recent years, the BDS-3 PPP-B2b of China and the Galileo High Accuracy Service (HAS) of the European Union (EU) have been transmitting real-time precise correction products via satellite-based broadcasting. PPP-B2b and HAS can realize the real-time precise positioning independent of network communication, which presents a new solution for wide-area real-time high-precision positioning.

Many scholars have evaluated the quality of precise corrections and the positioning performance of BDS-3 PPP-B2b service since its launch in 2020. The results indicated that the accuracy of satellite orbits and clock offsets was at the centimeter level and within 0.2 ns, respectively, and the satellite clock offsets presented a systematic bias. Moreover, the availability of PPP-B2b precise corrections was more

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than 90% in China, and the kinematic positioning accuracy of real-time PPP could reach centimeter level after convergence (Xu et al. 2021; Tao et al. 2021). Tang et al. (2022) provided a detailed description of the generation strategies for various PPP-B2b correction products, and conducted a comprehensive evaluation of the quality of PPP-B2b precise corrections. In addition, several researchers also focused on these aspects and derived comparable results (Wu et al. 2023; Ouyang et al. 2023). With regard to the systematic bias in PPP-B2b satellite clock offsets, Sun et al. (2023) derived its estimate by taking the post-processed precise ephemerides as a reference, and then the corrected PPP-B2b satellite clock offsets by the estimate were applied to the real-time PPP processing. Alternatively, Xu et al. (2023) introduced an additional parameter to compensate for the systematic bias in the real-time PPP processing. Both approaches significantly improved the convergence speed of real-time PPP.

The Galileo HAS Signal-In-Space (SIS) testing activities were conducted in the 2020–2022 timeframe. Fernandez-Hernandez et al. (2022) provided a detailed explanation of the message structure for Galileo HAS, including the organization and transmission mode of correction data, as well as how HAS uses Galileo monitoring stations and uplinks to provide services. HAS delivers precise correction products through the E6-B signal of Galileo satellites, and provides high-precision real-time PPP service to global users. The EU Agency for the Space Programme (EUSPA) officially announced that Galileo provided the initial HAS service on January 24, 2023. Since then, a number of scholars have conducted research on the availability, the quality of satellite orbit and clock corrections, and the positioning performance of Galileo HAS. Zhang et al. (2024b) evaluated the number of visible satellites and the service rate of HAS on a global scale, and the results showed that the globally average number of GPS and Galileo satellites with valid HAS corrections was 9.44 and 7.95, respectively, with the average service rate being 99.9% and 99.6% for GPS-only and Galileo-only PPP, respectively. Zhou et al. (2024) assessed the accuracy of HAS orbit and clock correction products, and analyzed the HAS-based positioning accuracy. The results indicated that the accuracy of orbit and clock corrections of Galileo was slightly higher than that of GPS, and both static and kinematic positioning of HAS could reach centimeter level. Similar results about the accuracy of HAS correction products and the HAS-based positioning performance were also derived by some other scholars (Naciri et al. 2023; Kan et al. 2024). In addition, receivers that support the E6 signal can directly receive the HAS correction message broadcast by Galileo satellites, but this is restricted by the baud rate of E6-B signal and the particularity of HAS encoding scheme (Fernandez-Hernandez et al. 2022). Therefore, some scholars developed open-source libraries or software such as HASlib, GHASP, and HASPPP to facilitate the decoding and integration of HAS corrections (Borio et al. 2023; Zhang et al. 2024a; Prol et al. 2024).

BDS-3 PPP-B2b can provide real-time PPP service in the Asia-Pacific region, while Galileo HAS is able to provide real-time PPP service for global users. Thus, the service areas of PPP-B2b and HAS overlap in the Asia-Pacific region. Moreover, PPP-B2b service supports BDS-3 and GPS constellations, while HAS service supports Galileo and GPS constellations. Therefore, there is potential for interoperability between PPP-B2b and HAS services regarding the supported satellite constellations. This study constructs a tightly integrated real-time PPP model with the joint use of PPP-B2b and HAS precise corrections in the Asia-Pacific region. For completeness, four real-time PPP strategies are analyzed and compared, including the stand-alone PPP-B2b strategy, stand-alone HAS strategy, and two PPP-B2b and HAS integration strategies. We first introduce the recovery of precise correction products and uniformity of datum. Next, the PPP-B2b and HAS tightly integrated real-time PPP model is constructed. Then, the results are presented and discussed. Finally, the main conclusions are summarized.

Recovery of precise correction products and uniformity of datum

The orbit and clock corrections of PPP-B2b refer to the legacy navigation message (LNAV) of GPS and the civil navigation message on B1C frequency (CNAV1) of BDS-3, while HAS provides orbit and clock corrections for LNAV of GPS and integrity navigation message (I/NAV) of Galileo. The decoded corrections are matched with the corresponding broadcast ephemerides to recover the real-time precise satellite orbits and clock offsets, that is:

$$\begin{cases}\tilde{O}_{B2b} = O_{B2b} - R_{rac2xyz}\delta O_{B2b} \\ \tilde{O}_{HAS} = O_{HAS} - R_{ntw2xyz}(\delta O_{HAS} + \delta\dot{O}_{HAS}\Delta t)\end{cases} \tag{1}$$

$$\begin{cases}\tilde{\mathrm{cdt}}^{s}_{B2b} = \mathrm{cdt}^{s}_{B2b} - \delta C_{B2b}/c \\ \tilde{\mathrm{cdt}}^{s}_{HAS} = \mathrm{cdt}^{s}_{HAS} + \delta C_{HAS}/c\end{cases} \tag{2}$$

where the subscripts B2b and HAS represent PPP-B2b and HAS corrections, respectively, $\tilde{O}$ and $\tilde{\mathrm{cdt}}^s$ are the recovered precise satellite positions and clock offsets, respectively, O and cdt^s are the satellite positions and clock offsets calculated by broadcast ephemerides, respectively, $R_{rac2xyz}$ and $R_{ntw2xyz}$ represent the transformation matrix from the radial, along-track and cross-track (RAC) coordinate system to the Earth-Centered Earth-Fixed (ECEF) coordinate system, both of which are calculated by the satellite position and velocity vectors derived from broadcast ephemerides, δO represents the corrections of satellite orbits, $\delta \dot{O}$ denotes the changing rates of orbit corrections, δC represents the corrections of satellite clock offsets, Δt is the time interval from the reference time of state-space representation (SSR) messages containing orbit corrections to the current time, and c is the speed of light in a vacuum.

Table 1 lists the datums of recovered real-time precise satellite orbits and clock offsets of PPP-B2b and HAS. The reference of GNSS code and carrier phase observations is the instantaneous antenna phase center (APC) of the satellite and the receiver. However, the orbit corrections provided by PPP-B2b refer to the L1/L2 ionospheric-free (IF) combined APC for GPS and the B3 APC for BDS-3, respectively, and the corresponding reference of orbit corrections provided by HAS is the L1 APC and E1 APC for GPS and Galileo, respectively. To ensure the consistency, the datums of recovered real-time precise satellite orbits of each satellite must be converted to the center of mass (CoM) by using the phase center offset (PCO) corrections in the ANTenna Exchange format (ANTEX) file provided by IGS, that is:

$$\tilde{O}_{COM} = \tilde{O}_{i,APC} - Tr \cdot PCO_i \tag{3}$$

where i denotes L1/L2 IF APC, B3 APC, L1 APC, or E1 APC, $\tilde{O}_{APC}$ and $\tilde{O}_{COM}$ represent the satellite orbits referring to the APC and CoM, respectively, Tr is a conversion matrix from the satellite body-fixed coordinate system to the ECEF coordinate system, and PCO denotes the PCO corrections. The reference of observations is transferred to the CoM by applying PCO and phase center variation (PCV) corrections for the consistency of PPP processing.

As shown in Table 1, the clock offset datum of GPS and BDS-3 in PPP-B2b products is L1P/L2P IF combination and B3I, respectively, while that of GPS and Galileo in HAS products is the IF combination with L1 C/A and L2P, and E1 and E5B, respectively. For real-time PPP processing with multiple frequencies, the PPP-B2b and HAS also provide the code observable-specific signal bias (OSB) correction products. Hence, we just need to directly apply the OSB corrections to the code observations on the required frequency, so as to be consistent with the reference of the recovered real-time precise satellite clock offsets. It should be noted that the sign of the PPP-B2b and HAS correction formulas is opposite when using the augmentation services for OSB corrections (EUSPA 2022). For convenience, the single-frequency reference is denoted by m, and the dual-frequency reference is denoted by m and n. Thus, the specific components of the recovered real-time precise satellite clock offsets can be expressed as:

$$\begin{cases} c\tilde{d}t^s_m = cdt^s - d^s_m \\ c\tilde{d}t^s_{IF,mn} = cdt^s - d^s_{IF,mn} \end{cases} \tag{4}$$

$$d^s_{IF,mn} = a_{mn,m} \cdot d^s_m + a_{mn,n} \cdot d^s_n \tag{5}$$

with $a_{mn,m} = f_m^2/(f_m^2 - f_n^2)$ and $a_{mn,n} = -f_n^2/(f_m^2 - f_n^2)$ where $a_{mn,m}$ and $a_{mn,n}$ are the dual-frequency IF combination coefficients, f is the carrier frequency, d^s is the satellite code hardware delay, $c\tilde{d}t^s_m$ denotes the precise satellite clock offset with a single-frequency datum, and $c\tilde{d}t^s_{IF,mn}$ denotes the precise satellite clock offset with a dual-frequency IF combined datum.

The raw code observations contain the satellite code hardware delay on a single frequency, which needs to be corrected by OSB to make them consistent with the satellite code hardware delay included in the precise satellite clock offsets. The BDS-3 code observations can be corrected by OSB from PPP-B2b according to the following formula:

$$\bar{P}_j = P'_j + d^s_j - OSB_{m,j} = P'_j + d^s_m \tag{6}$$

where j represents the frequency, OSB represents the code OSB corrections, $\overline{P}$ represents the code observations corrected by OSB, and P' represents the code observations without the satellite code hardware delay. It should be noted that the PPP-B2b product does not contain the OSB corrections of GPS. Since the GPS satellite clock offsets from PPP-B2b product take a L1/L2 dual-frequency IF combined datum, the inconsistency of satellite code hardware delay between code observations and satellite clock offsets will be absorbed by the ionospheric delay parameter when employing a L1/L2 dual-frequency uncombined observation model.

The Galileo and GPS code observations can be corrected by OSB from HAS according to the following formula:

$$\bar{P}_j = P'_j + d^s_j + OSB_{IF,mn,j} = P'_j + d^s_{IF,mn} \tag{7}$$

Table 1 Datums of recovered real-time precise satellite orbits and clock offsets of PPP-B2b and HAS

Items	Systems	Datums	
		PPP-B2b	HAS
Satellite orbits	GPS	L1 and L2 IF APC	L1 APC
	BDS-3	B3 APC	–
	Galileo	–	E1 APC
Clock offsets	GPS	L1P and L2P IF	L1 C/A and L2P IF
	BDS-3	B3I	–
	Galileo	–	E1 and E5B IF

PPP-B2b and HAS tightly integrated real-time PPP model

Since both PPP-B2b and HAS products support GPS, for ease of expression, the GPS constellation taking PPP-B2b corrections is represented as GPS B2b, and the GPS constellation taking HAS corrections is represented as GPS HAS. The code and phase observations can be expressed as follows:

$$\begin{cases} P_{r,j}^{s,sys} = \rho_r^{s,sys} + c\mathrm{d}t_r^{sys} - c\mathrm{d}t^{s,sys} \\ +\gamma_j^{sys} \cdot I_r^{s,sys} + T_r^{s,sys} + d_{r,j}^{sys} + d_j^{s,sys} + e_{r,j}^{s,sys} \\ L_{r,j}^{s,sys} = \rho_r^{s,sys} + c\mathrm{d}t_r^{sys} - c\mathrm{d}t^{s,sys} - \gamma_j^{sys} \cdot I_r^{s,sys} \\ +T_r^{s,sys} + N_{r,j}^{s,sys} + b_{r,j}^{sys} + b_j^{s,sys} + \varepsilon_{r,j}^{s,sys} \end{cases} \tag{8}$$

where sys represents GPS B2b, GPS HAS, BDS-3 or Galileo, *r* and *s* represent the receiver and satellite, respectively, *j* is the frequency, *P* represents code observations, *L* represents carrier phase observations, ρ is the geometric distance between satellite and receiver, $c\mathrm{d}t_r$ and $c\mathrm{d}t^s$ represent the receiver and satellite clock offsets, respectively, *I* represents the slant ionospheric delay on the first frequency, γ is the ionospheric delay factor ($\gamma_j^{sys} = (f_1^{sys}/f_j^{sys})^2$), *f* is the carrier frequency, *T* represents the slant tropospheric delay, d_r and d^s denote the receiver and satellite code hardware delay, respectively, *N* is the phase ambiguity, b_r and b^s denote the receiver and satellite phase hardware delay, respectively, and *e* and ε represent the measurement noises of code and carrier phase observations (including unmodeled errors such as multipath effects), respectively.

The datums of satellite clock offsets of GPS B2b are frequently switched because the PPP-B2b products are derived from the regional network solutions. Therefore, a separate receiver clock offset parameter should be estimated for each satellite system. Alternatively, the inter-system bias (ISB) parameter should be estimated as white noise process to absorb the influence of switched datums. The former method is adopted in this study. It should be noted that there is an initial systematic bias in the recovered PPP-B2b precise satellite clock offsets for each satellite, which can be absorbed by the phase ambiguity parameter and does not affect the carrier phase observations. However, the initial systematic bias will impact the accuracy of code observations. Therefore, a compensating parameter is introduced into the code observation equation for each satellite to mitigate the effects of the initial systematic bias included in satellite clock offsets for GPS B2b and BDS-3 constellations. Taking the final satellite clock product as the reference, the errors of PPP-B2b satellite clock offsets exhibit relatively stable characteristics within each continuous arc. Thus, the compensating parameter for the initial systematic bias is estimated as random-walk process. After applying the recovered real-time precise satellite orbits and clock offsets as well as the OSB corrections, the PPP-B2b and HAS tightly integrated real-time PPP dual-frequency uncombined observation model can be expressed as:

$$\begin{cases} p_{r,1}^{s,sys} = \mu_r^{s,sys} \cdot X_r + c\bar{\mathrm{d}}t_r^{sys} + \gamma_1^{sys} \cdot \bar{I}_r^{s,sys} \\ \quad +\alpha_r^{s,sys} \cdot Z_r + \theta^{sys} \cdot D^{s,sys} \\ p_{r,2}^{s,sys} = \mu_r^{s,sys} \cdot X_r + c\bar{\mathrm{d}}t_r^{sys} + \gamma_2^{sys} \cdot \bar{I}_r^{s,sys} \\ \quad +\alpha_r^{s,sys} \cdot Z_r + \theta^{sys} \cdot D^{s,sys} \\ l_{r,1}^{s,sys} = \mu_r^{s,sys} \cdot X_r + c\bar{\mathrm{d}}t_r^{sys} - \gamma_1^{sys} \cdot \bar{I}_r^{s,sys} \\ \quad +\alpha_r^{s,sys} \cdot Z_r + \bar{N}_{r,1}^{s,sys} \\ l_{r,2}^{s,sys} = \mu_r^{s,sys} \cdot X_r + c\bar{\mathrm{d}}t_r^{sys} - \gamma_2^{sys} \cdot \bar{I}_r^{s,sys} \\ \quad +\alpha_r^{s,sys} \cdot Z_r + \bar{N}_{r,2}^{s,sys} \end{cases} \tag{9}$$

$$\theta^{sys} = \begin{cases} 0, \; sys = \text{GPS HAS or Galileo} \\ 1, \; sys = \text{GPS B2b or BDS-3} \end{cases} \tag{10}$$

$$\begin{cases} c\bar{\mathrm{d}}t_r^{sys} = c\mathrm{d}t_r^{sys} + a_{12,1}^{sys} \cdot d_{r,1}^{sys} + a_{12,2}^{sys} \cdot d_{r,2}^{sys} \\ \bar{I}_r^{s,sys} = I_r^{s,sys} + a_{12,2}^{sys} \cdot (d_{r,1}^{sys} - d_{r,2}^{sys}) + \eta^{sys} \cdot d_{ion}^{s,sys} \\ \bar{N}_{r,1}^{s,sys} = N_{r,1}^{s,sys} + b_{r,1}^{sys} + b_1^{s,sys} - (a_{12,1}^{sys} - \gamma_1^{sys} \cdot a_{12,2}^{sys}) \cdot d_{r,1}^{sys} \\ \quad - a_{12,2}^{sys} \cdot (1 + \gamma_1^{sys}) \cdot d_{r,2}^{sys} - d_{pre}^{s,sys} + \eta^{sys} \cdot \gamma_1^{sys} \cdot d_{ion}^{s,sys} \\ \bar{N}_{r,2}^{s,sys} = N_{r,2}^{s,sys} + b_{r,2}^{sys} + b_2^{s,sys} - (a_{12,1}^{sys} - \gamma_2^{sys} \cdot a_{12,2}^{sys}) \cdot d_{r,1}^{sys} \\ \quad - a_{12,2}^{sys} \cdot (1 + \gamma_2^{sys}) \cdot d_{r,2}^{sys} - d_{pre}^{s,sys} + \eta^{sys} \cdot \gamma_2^{sys} \cdot d_{ion}^{s,sys} \end{cases} \tag{11}$$

$$d_{pre}^{s,sys} = \begin{cases} d_m^{s,sys}, \quad sys = \text{BDS-3} \\ a_{mn,m}^{sys} \cdot d_m^{s,sys} + a_{mn,n}^{sys} \cdot d_n^{s,sys}, sys = \text{others} \end{cases} \tag{12}$$

$$\eta^{sys} = \begin{cases} 1, \quad sys = \text{GPS B2b} \\ 0, \quad sys = \text{others} \end{cases} \tag{13}$$

$$d_{ion}^{s,sys} = a_{mn,n}^{sys} \cdot (d_m^{s,sys} - d_n^{s,sys}) \tag{14}$$

with $a_{12,1}^{sys} = (f_1^{sys})^2/((f_1^{sys})^2 - (f_2^{sys})^2)$ and $a_{12,2}^{sys} = -(f_2^{sys})^2/((f_1^{sys})^2 - (f_2^{sys})^2)$ where p and l denote the observed-minus-computed (OMC) code and phase observables, respectively, μ denotes the unit vector in the line of sight direction from the receiver to the satellite, X denotes the three-dimensional (3D) coordinates of the receiver, $c\bar{\mathrm{d}}t_r$ denotes the receiver clock offset estimate that absorbs the receiver code hardware delay in the form of the dual-frequency IF combination, $\bar{I}$ denotes the ionospheric delay estimate that absorbs the code hardware delay at the receiver as well as the code hardware delay at the satellite (only for GPS B2b), *Z* denotes the tropospheric zenith wet delay (ZWD), α denotes the wet mapping function, $\bar{N}$ denotes the phase ambiguity estimate that absorbs the code and phase hardware delay at the satellite and receiver, and *D* denotes the compensating parameter for the initial systematic bias of PPP-B2b satellite clock offsets. In Eq. (11), $d_{pre}^{s,sys}$ denotes the satellite code hardware delay absorbed by the phase ambiguity parameter, which is introduced when applying the PPP-B2b or HAS precise satellite clock offsets, and is consistent with their reference (see

Eq. (4)). In Eq. (11), $d_{ion}^{s,sys}$ denotes satellite code hardware delay absorbed by the ionospheric delay parameter and then propagated to the phase ambiguity parameter (only for GPS B2b), which is introduced by the inconsistency of satellite code hardware delay between code observations and satellite clock offsets. To remove the rank deficiency caused by the introduction of compensating parameter for the initial systematic bias for each satellite within a constellation in Eq. (9), a separate pseudo-observation equation of the compensating parameter with a pseudo-observable of zero for a selected satellite for each of GPS B2b and BDS-3 constellations can be introduced into the observation model.

In view that the pseudo-observable is introduced as a datum to eliminate the rank deficiency, a much smaller magnitude should be assigned to its noise. In this study, the variance of the pseudo-observable is set to 1×10^{-6} m^2. There is a strong correlation between the receiver clock offset parameter and numerous compensating parameters. By assigning a numerical value of zero to the compensating parameter of the selected satellite, the receiver clock offset estimates actually absorb the initial systematic bias of the selected satellite, while the estimated compensating parameter for other satellites is essentially the difference of initial systematic bias between these satellites and the selected satellite. Considering the regional service nature of BDS-3 PPP-B2b, even when the selected satellite exits the service coverage, the introduced datum will propagate backward to subsequent epochs because the compensating parameter is modeled as random-walk process. During satellite ingress events, compensating parameters for the newly visible satellites require initialization, whereas compensating parameters for all satellites must undergo re-initialization upon re-entry of the selected satellite into the service area.

Since the estimator adopts the Kalman filter, there is predicted information for the parameters with a strongly constrained dynamic model. Thus, as the compensating parameter is estimated as random-walk process, the introduction of a compensating parameter into the code observation equation for each satellite of GPS B2b and BDS-3 constellations will not reduce the redundancy of the real-time PPP model. Actually, the redundancy of the model approximates the difference between the number of observations and the number of parameters without strong constraints, such as receiver clock offset parameters.

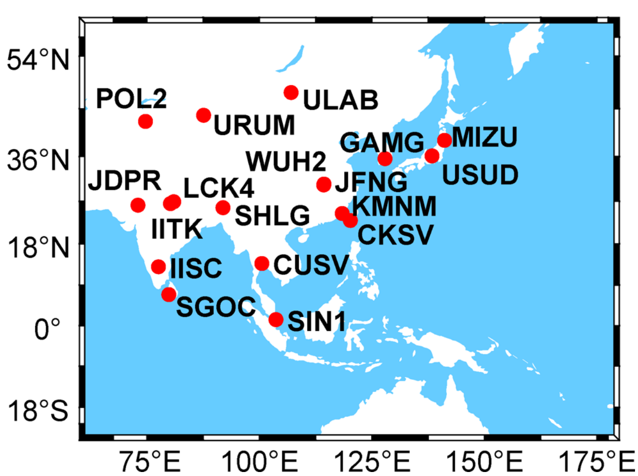

Fig. 1 Geographical distribution of 18 MGEX stations located in the Asia-Pacific region

Results and discussion

The datasets and processing strategies are first introduced. Next, the availability is analyzed. Finally, the position solutions are provided.

Datasets and processing strategies

In this study, the datasets on day of year (DOY) 180–186 of 2024 from 18 Multi-GNSS Experimental (MGEX) stations located in the Asia-Pacific region are employed to evaluate the performance of PPP-B2b and HAS tightly integrated real-time PPP. Due to the absence of observations at certain stations on specific days (i.e., JDPR, LCK4 and IITK stations on DOY 185 of 2024 and SGOC station on DOY 180 of 2024), the analysis ultimately involves a total of 122 sets of daily observations. The geographical distribution of the selected stations is shown in Fig. 1. The stations are chosen according to three principles: (1) coverage within the PPP-B2b service area (75°E–135°E, 10°N–55°N) with minor boundary extensions of several degrees, (2) support for simultaneous tracking of BDS-3, GPS, and Galileo signals, and (3) quasi-uniform spatial distribution. As for the analysis period, while it is arbitrarily chosen, we adopt a continuous 7-day period after considering the orbital repeat period and the computational burden.

For comparison, four real-time PPP strategies are analyzed, including the strategy with BDS-3/GPS from PPP-B2b (B2b C/G), the strategy with Galileo/GPS from HAS (HAS E/G), the strategy with BDS-3/GPS from PPP-B2b and Galileo from HAS (B2b C/G + HAS E), and the strategy with Galileo/GPS from HAS and BDS-3 from PPP-B2b (HAS E/G + B2b C). The real-time PPP processing is carried out with the strategies summarized in Table 2.

Availability

The availability analysis presented in this study is concentrated on the Asia-Pacific region with specific geographical ranges from latitude 0° to 60°N and longitude 60°E to 180°. The service rates, the number of visible satellites, and the position dilution of precision (PDOP) for four real-time PPP

Table 2 Processing strategies for real-time PPP

Items	Strategies
Frequencies	BDS-3: B1/B3 GPS: L1/L2 Galileo: E1/E5A
Positioning mode	Kinematic
Weighting method	Elevation-dependent weights
Sampling rate	30 s
Elevation cut-off angle	10°
PCO/PCV	PPP-B2b: IGS20 atx file HAS: IGS14 atx file
Phase wind-up effect	Corrected
Relativistic effect	Corrected
Station displacement	Corrected with IERS Convention 2010
Satellite orbit/clock/code bias	Broadcast ephemerides with corresponding augmentation products
Receiver clock	A separate receiver clock offset parameter is estimated for each satellite system, and modeled as white noise process (10^4 m^2)
Ionospheric delay	Estimated as random-walk process (10^{-4} m^2/s)
Tropospheric delay	Dry component: Saastamoinen model Wet component: mapping slant delays to zenith delays with global mapping function (GMF) and estimated as random-walk process (10^{-8} m^2/s)
Phase ambiguity	Estimated as float constants
Compensating parameter for initial systematic bias	Estimated as random-walk process (10^{-10} m^2/s) over each continuous arc

strategies (namely B2b C/G, B2b C/G + HAS E, HAS E/G, and HAS E/G + B2b C) are analyzed. To ensure the consistency with the positioning strategies, the sampling interval is set to 30 s, and the cut-off elevation angle is set to 10°. The research area is divided into 60 × 60 grids with an interval of 2° in longitude and 1° in latitude, and it is assumed that a virtual receiver at an altitude of 25 m is placed in the center of each grid (Yang et al. 2011). Subsequently, the relative geometries between each satellite and the virtual receiver are analyzed to evaluate the availability of four real-time PPP strategies by combining the known receiver positions and the calculated satellite positions.

When there are at least four visible satellites and the PDOP values are less than 30 at a certain epoch, the position determination at this epoch is considered to be feasible. In this study, the service rate is defined as the percentage of the number of epochs with feasible position solutions over the total number of epochs. The service rate for each grid is calculated. The results indicate that all four real-time PPP strategies can achieve a service rate of 100% in the Asia-Pacific region, which means that even the stand-alone PPP-B2b service and the stand-alone HAS service can provide continuous positioning services for users in the Asia-Pacific region.

Figure 2 illustrates the geographical distribution of the average number of visible satellites over seven days from DOY 180 to 186 of 2024 for four real-time PPP strategies.

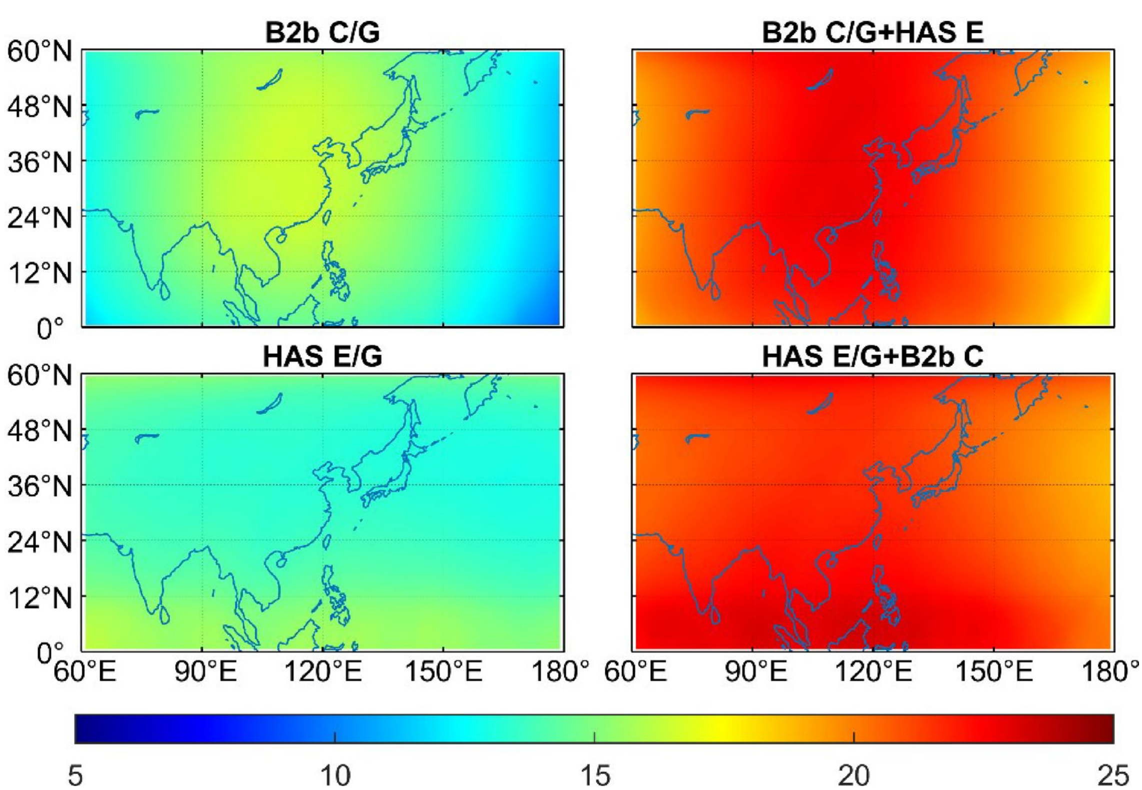

Fig. 2 Geographical distribution of the average number of visible satellites for four real-time PPP strategies

The number of visible satellites under the B2b C/G strategy is relatively larger in the mainland of China and its coastal regions, especially within the range of 12°N to 48°N and 90°E to 130°E, with the corresponding satellite numbers ranging from 14.9 to 16.5. In addition, the visible satellite numbers present a decreasing trend towards surrounding areas in a circular pattern, with the lowest number in the Pacific region being 9.3. After integrating with HAS Galileo, the number of visible satellites increases significantly, ranging from 16.6 to 22.9, and the distribution characteristics of satellite numbers are consistent with those of the B2b C/G strategy (i.e., more visible satellites in the central regions and less visible satellites in the peripheral regions). The number of visible satellites under the HAS E/G strategy is mainly concentrated in the areas spanning 0° to 12°N and 50°N to 60°N, with the corresponding satellite numbers ranging from 13.2 to 16.2. However, in the mainland of China and its coastal regions, the number of visible satellites under the HAS E/G strategy is relatively smaller, with a varying range of 12.9 to 14.8. After integrating with PPP-B2b BDS-3, the distribution characteristics of the satellite numbers are consistent with those of the HAS E/G strategy, and there are less visible satellites (ranging from 18.7 to 21.2) in the regions spanning 12°N to 50°N (especially in the regions of 160°E to 180°), while the corresponding numbers increase to 23.3 in other regions.

Figure 3 displays the geographical distribution of average PDOP values over the entire analysis period for four real-time PPP strategies. The B2b C/G+HAS E strategy integrating HAS Galileo presents lower PODP values in contrast to the B2b C/G strategy that only employs PPP-B2b products. Similarly, the HAS E/G+B2b C strategy that integrates PPP-B2b BDS-3 also exhibits lower PODP values compared with the HAS E/G strategy. The PDOP values of the B2b C/G and B2b C/G+HAS E strategies are relatively smaller within the range of 80°E to 140°E, particularly in the mainland of China, with the minimum values of 1.28 and 1.06, respectively. In the areas of 60°E to 80°E and 140°E to 180°, the PDOP values of B2b C/G and B2b C/G+HAS E strategies gradually increase towards both sides, ranging from 1.39 to 3.01 and 1.11 to 1.41, respectively. For the other two strategies, the distribution of PDOP values is relatively even, ranging from 1.35 to 1.62 for HAS E/G strategy and from 1.07 to 1.25 for HAS E/G+B2b C strategy. After an integration with PPP-B2b BDS-3, the HAS E/G+B2b C PDOP values within the latitude range from 12°N to 40°N are still slightly higher than those in other regions, especially in the longitude range from 150°E to 180°.

Position solutions

Figures 4 and 5 depict the epoch-wise positioning errors for four real-time PPP strategies (i.e., B2b C/G, B2b C/G+HAS

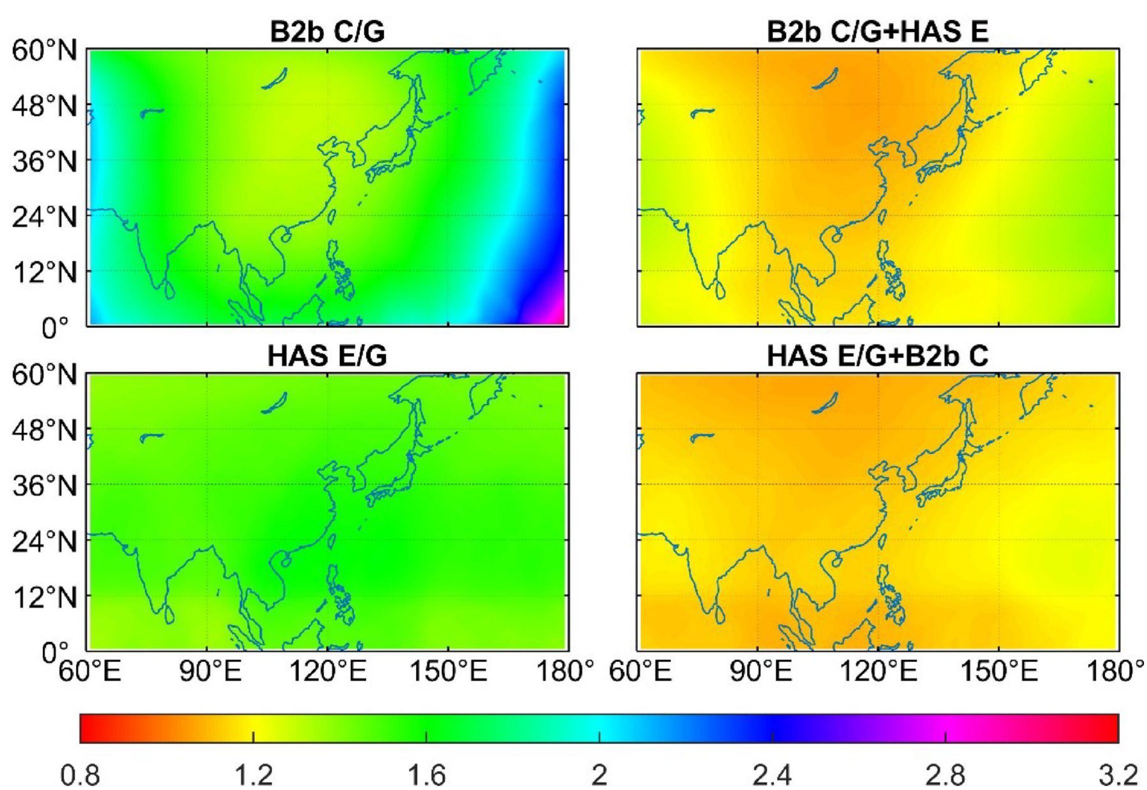

Fig. 3 Geographical distribution of average PDOP values for four real-time PPP strategies

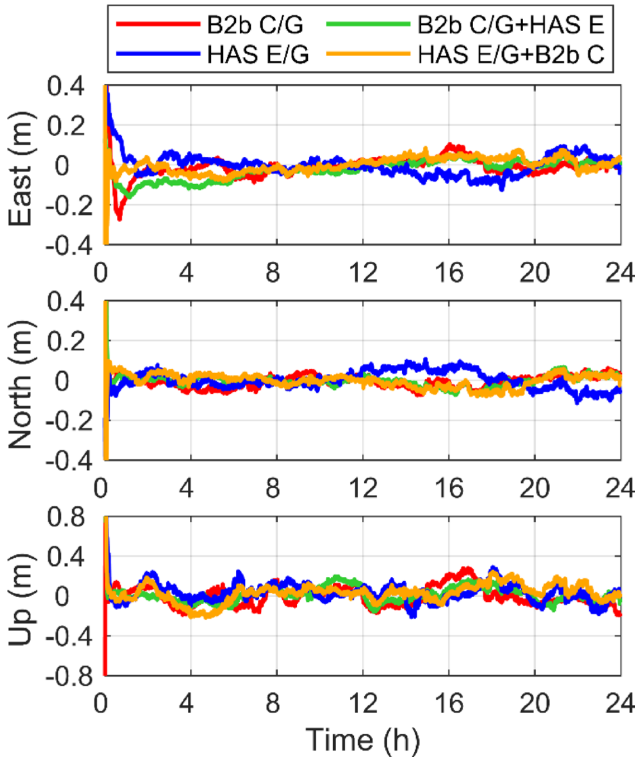

Fig. 4 Epoch-wise positioning errors for four real-time PPP strategies at station JDPR on DOY 186 of 2024

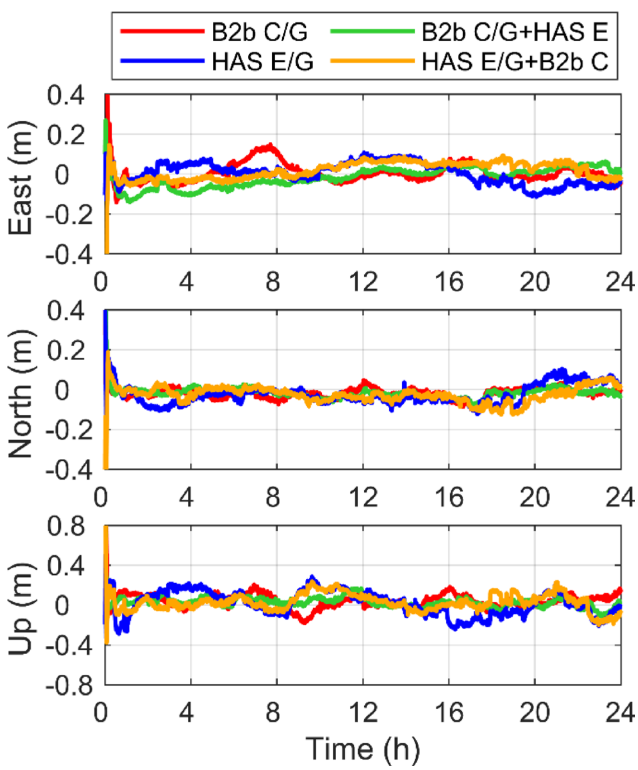

Fig. 5 Epoch-wise positioning errors for four real-time PPP strategies at station POL2 on DOY 186 of 2024

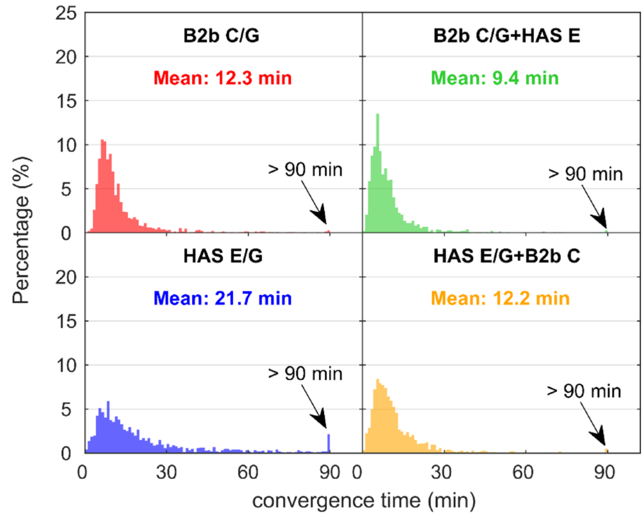

Fig. 6 Distribution of convergence time for four real-time PPP strategies

E, HAS E/G, and HAS E/G+B2b C) at stations JDPR and POL2 on DOY 186 of 2024, respectively. As can be seen from the two figures, the position solutions of all the four real-time PPP strategies can quickly converge to the target accuracy of PPP-B2b and HAS services (i.e., 20 cm in horizontal direction and 40 cm in vertical direction). The positioning accuracy in the east and north directions is usually better than 10 cm after convergence, while the corresponding accuracy in the up direction is usually better than 20 cm. It should be noted that the stability of position solutions of HAS E/G strategy is inferior to that of the other three strategies after the position filter converges.

To evaluate the convergence time of four real-time PPP strategies, the observations from 18 stations in the Asia-Pacific region on seven consecutive days are employed. In PPP processing, the estimator is reset every two hours to obtain the position solutions, and thus a total of 1464 sets of 2-hour position solutions are counted. When the horizontal positioning errors are less than 20 cm and the vertical positioning errors are less than 40 cm within 10 consecutive epochs (i.e., 5 min) (CSNO 2021; EUSPA 2023), the position solutions are considered to have converged. The distribution of convergence time is plotted in Fig. 6. The proportion of real-time PPP solutions with a convergence time longer than 90 min is represented as a 90-minute statistic in Fig. 6, and the average convergence time of each strategy is also provided in each panel. It can be seen that the B2b C/G+HAS E strategy achieves the shortest convergence time of 9.4 min on average, followed by B2b C/G and HAS E/G+B2b C strategies with an average convergence time of 12.3 and 12.2 min, respectively. In contrast, the HAS E/G strategy exhibits the slowest convergence performance with an average convergence time of 21.7 min. The real-time PPP solutions with a convergence time less

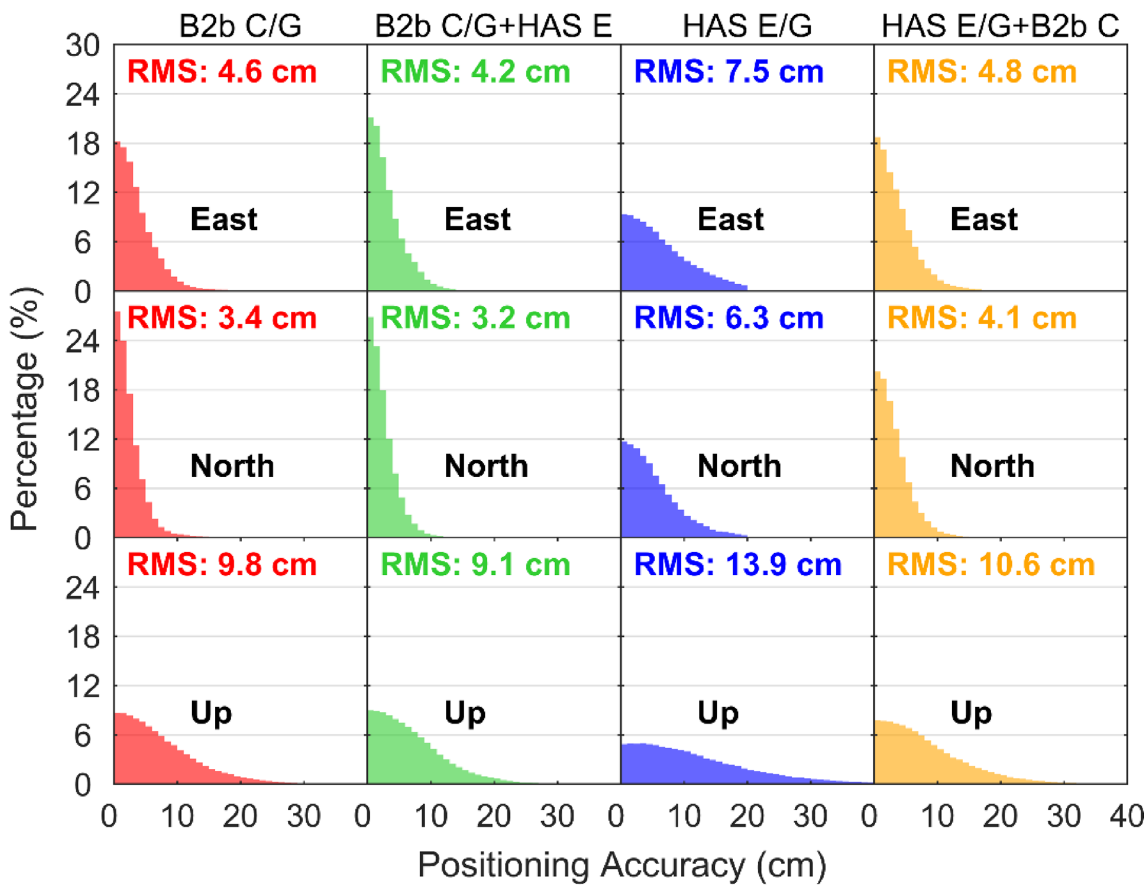

Fig. 7 Distribution of positioning accuracy for four real-time PPP strategies

than 10/20 min account for 59.6%/87.4%, 72.4%/92.6%, 35.1%/64.3%, and 58.6%/86.9% for the B2b C/G, B2b C/G+HAS E, HAS E/G, and HAS E/G+B2b C strategies, respectively. The average convergence time of B2b C/G+HAS E strategy is shortened by 2.9 min compared with the B2b C/G strategy, with an improvement of 24%, which can be reflected in the increase of the proportion of position solutions with a convergence time shorter than 15 min. Compared with the HAS E/G strategy, the average convergence time of HAS E/G+B2b C strategy is shortened by 9.5 min and the improvement rate is 44%, which is reflected not only in the increased proportion of position solutions with a convergence time less than 15 min, but also in the significantly reduced proportion of real-time PPP solutions with a convergence time over 90 min.

To evaluate the kinematic positioning accuracy of four real-time PPP strategies, the observations from 18 stations in the Asia-Pacific region on seven consecutive days are processed every 24 h. To avoid the influence of position solutions in the converging stage, the results within the first two hours of each day are removed. Figure 7 shows the distribution of the absolute value of epoch-wise positioning errors for all available epochs. The root mean square (RMS) statistics of positioning errors over all available epochs for each real-time PPP strategy are calculated and shown in each panel. The results indicate that the B2b C/G+HAS E strategy can achieve the highest positioning accuracy, followed by B2b C/G and HAS E/G+B2b C strategies, while the HAS E/G strategy has the lowest positioning accuracy. The real-time PPP solutions with a positioning accuracy better than 10/10/20 cm in the east/north/up directions account for 96.2%/98.4%/95.2%, 97.5%/99.3%/96.7%, 80.3%/88.3%/84.1%, and 95.3%/97.7%/93.5% for the B2b C/G, B2b C/G+HAS E, HAS E/G, and HAS E/G+B2b C strategies, respectively. Compared with the B2b C/G strategy, the positioning accuracy of B2b C/G+HAS E strategy is improved by 9%/6%/7% in the east/north/up directions, respectively, with the RMS statistics being 4.2/3.2/9.1 cm and 4.6/3.4/9.8 cm for the two strategies, respectively. Similarly, compared with the HAS E/G strategy, the HAS E/G+B2b C strategy improves the positioning accuracy by 36%/35%/24% in the three directions, respectively, and the corresponding RMS statistics are 4.8/4.1/10.6 cm and 7.5/6.3/13.9 cm for the two strategies, respectively.

By comparing the convergence time and positioning accuracy among the four real-time PPP strategies, it can be found that the positioning performance of the HAS E/G strategy is significantly worse than that of the B2b C/G strategy. Even after the introduction of BDS-3 observations with PPP-B2b service, the positioning performance of the HAS E/G+B2b C strategy still remains inferior to that of the B2b C/G strategy. It is indicated that the positioning performance of HAS service is worse than that of PPP-B2b service in the Asia-Pacific region. This is because there are few monitoring stations used to generate the precise correction products of HAS service in the Asia-Pacific region, and in fact, the HAS service does not commit to guaranteeing the positioning performance in the Asia-Pacific region (EUSPA 2023). Despite this, both the convergence time and positioning accuracy can be significantly improved by integrating the HAS-based Galileo observations into the PPP-B2b service.

Conclusions

Considering that the service areas of BDS-3 PPP-B2b and Galileo HAS overlap in the Asia-Pacific region, and there is potential for interoperability between the two services in terms of supported satellite constellations, a tightly integrated real-time PPP model with PPP-B2b and HAS for the Asia-Pacific region is constructed in this study. Four real-time PPP strategies, including B2b C/G, B2b C/G+HAS E, HAS E/G, and HAS E/G+B2b C, are used for comparative analysis in terms of availability, convergence time, and positioning accuracy.

Regarding the availability in the Asia-Pacific region, among the four real-time PPP strategies, the number of visible satellites of B2b C/G+HAS E and HAS E/G+B2b C strategies is at a similar level, ranging from about 16 to 24, and the distribution characteristics of PDOP values are closely related to the number of visible satellites, with the corresponding statistics being 1.06 to 1.41 and 1.07 to 1.25 for the two strategies, respectively. The B2b C/G strategy has the smallest number of visible satellites in the Pacific region, with the minimum value being 9.3, and the PDOP value can reach as high as 3.01. By contrast, in the mainland of China and its coastal regions, the number of visible satellites is increased to 14.9–16.5, and the corresponding PDOP value is relatively smaller, with a minimum value of 1.28. For the HAS E/G strategy, the number of visible satellites varies within a range of 12.9 to 16.2, and the PDOP values fluctuate between 1.35 and 1.62. In the Asia-Pacific region, all four real-time PPP strategies can achieve a service rate of 100%.

The convergence time of B2b C/G+HAS E strategy is the shortest at 9.4 min, while the B2b C/G and HAS E/G+B2b C strategies have a similar convergence time of 12.3 and 12.2 min, respectively. By contrast, the HAS E/G strategy has the longest convergence time at 21.7 min. Compared with the B2b C/G strategy, the convergence time of the B2b C/G+HAS E strategy is shortened by 24%, while the convergence time of the HAS E/G+B2b C strategy is shortened by 44% over the HAS E/G strategy. As for the positioning accuracy, the comparative results are similar to those of convergence time. The B2b C/G+HAS E strategy performs the best, followed by B2b C/G and HAS E/G+B2b C strategies, while the HAS E/G strategy ranks the lowest in the positioning accuracy. The positioning accuracy of the B2b C/G strategy can reach 4.6, 3.4 and 9.8 cm in the east, north and up directions, respectively, whereas the positioning accuracy of the B2b C/G+HAS E strategy can be improved by 9%, 6% and 7% to 4.2, 3.2 and 9.1 cm over the B2b C/G strategy in the three directions, respectively. The positioning accuracy of the HAS E/G strategy can reach 7.5, 6.3 and 13.9 cm in the three directions, respectively, while the positioning accuracy of the HAS E/G+B2b C strategy can be improved by 36%, 35% and 24% compared with the HAS E/G strategy in the three directions, reaching 4.8, 4.1 and 10.6 cm, respectively.

In summary, the tightly integrated real-time PPP with PPP-B2b and HAS in the Asia-Pacific region can significantly shorten the convergence time and improve the positioning accuracy compared with a stand-alone augmentation service, which verifies the effectiveness of the proposed model. However, the precise correction products of HAS service are generated by few monitoring stations in the Asia-Pacific region, and the HAS service does not commit to guaranteeing the positioning performance in the Asia-Pacific region. Therefore, the two HAS-based strategies, namely using the stand-alone HAS service and integrating the PPP-B2b BDS-3 observations into the HAS service, may not be as effective as expected in this region. It is expected that the positioning performance of tightly integrated real-time PPP with the two augmentation services will be further improved as their service levels develop in the future.

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Data availability The datasets at selected MGEX stations can be obtained from http://igs.gnsswhu.cn.

Declarations

Ethics approval and consent to participate Not applicable.

Competing interests The authors declare no competing interests.

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